

Mikoyan–Guryevich MiG-21

“It was superb to fly, tough, simple, and easy to build in large numbers...”

IVAN RENDALL
COMMENTING ON THE MiG-21
IN ROLLING THUNDER

THE MiG-21, known to NATO by the codename “Fishbed,” is an aircraft that benefits from a design focused on limited, attainable objectives. It was conceived in the aftermath of the Korean War, when the Soviet military decided they needed a new-generation short-range interceptor and air-superiority fighter. The aircraft had to be fast – capable of flying at Mach 2 – and maneuverable, with all other features sacrificed to high performance. It also needed to be simple, reliable, easy to maintain, and cheap enough to be manufactured in large numbers.

The design produced by the Mikoyan and Gurevich bureau was a “no frills,” stripped-down, classic dogfighter and bomber-killer. In 1959, at the same time as the US was developing the F-4 Phantom – which was heavy enough to require two engines and needed an Electronic Warfare Officer to operate its array of electronic gadgetry – the MiG-21

emerged as a single-seat, single-engine, lightweight fighter, with a simple radar, two heat-seeking missiles, and a cannon. When F-4 pilots first encountered MiG-21s over North Vietnam, their craft’s advanced electronics and extra engine power did not necessarily translate into combat victories. In fact the MiG was more nimble and tighter in a high-speed turn, and its gun gave a definite advantage over the US fighters, which initially did not feature a gun.

Over the years, the MiG-21 evolved away from its original lightness and simplicity. Later models had more sophisticated radar and an extra fuel tank to give the aircraft a greater range. The engine was modified to allow the airplane to carry more missiles as well as the extra fuel load. But the virtues of cheapness, reliability, and high performance remained. Over 13,000 MiG-21s were produced. They went into service with air forces around the world and saw action not only in Vietnam but also in other areas, such as the Middle East. Many were still operational at the start of the third millennium.

POPULAR CRAFT

Since it was first produced, the MiG-21 has been used by over 50 air forces and has seen service in at least 30 wars. The airplane shown here is a Yugoslavian Air Force MiG-21UTI, taking off from an airstrip in Kosovo, 1999.



Faring for tailplane actuator

Tailpipe of engine afterburner

Engine bay venting air intake

Communications antenna

Ventral fin

Speed brake

Main undercarriage

PREPARING FOR ACTION

A group of Soviet pilots rush toward a line of waiting MiG-21s during a training scramble in Moscow, July 1965.

